

English-Only Confidence Retry Experiment

PP-LCNet_x1_0_doc_ori — Set 1 (Clean) vs Set 2 (Aug + Messify)

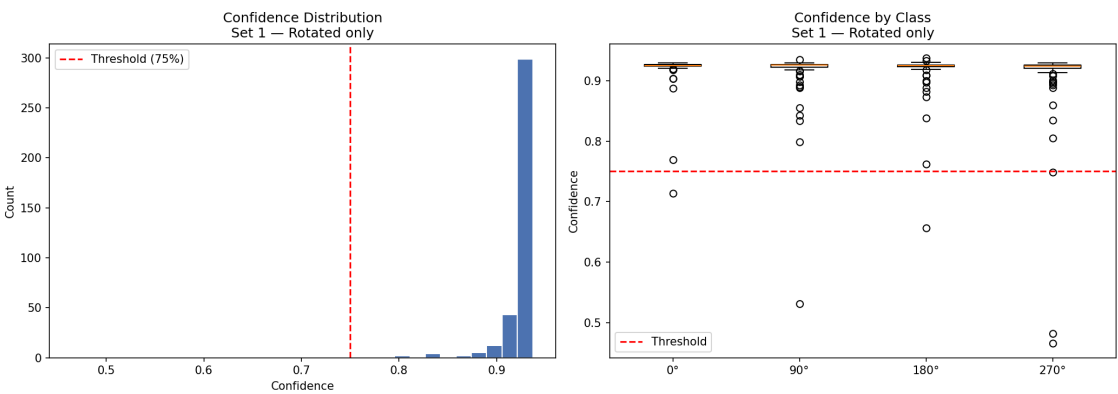
Threshold: 75% | 7 PDFs, 94 pages, 376 images per set | ONNX Runtime

1. Objective

For each English-only image where the model confidence falls below 75%, the image is rotated to all 4 orientations and inference is re-run on each. The rotation yielding the highest confidence becomes the final prediction. This experiment runs on both the clean set (Set 1) and the augmented/messy set (Set 2) to isolate how degradation affects the retry strategy.

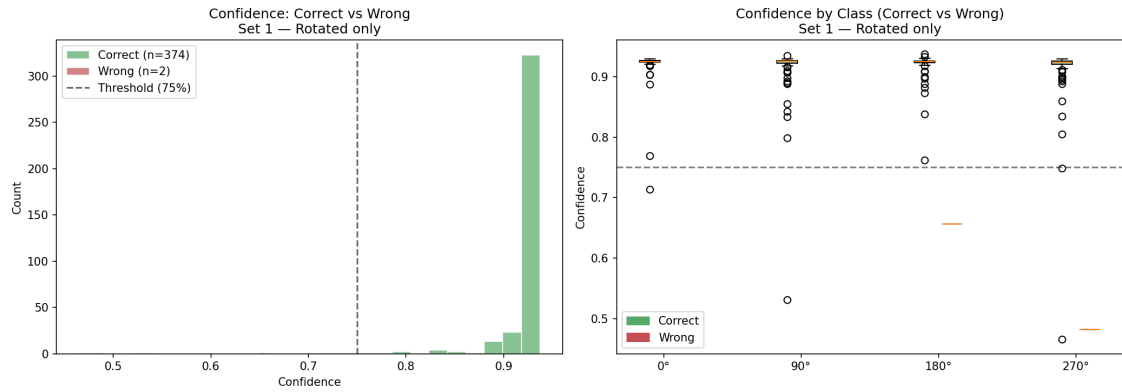
2. Set 1 — Rotated only

Metric	Value
Images	376
Overall accuracy	99.47%
Avg confidence	91.5%
Uncertain (<75%)	6 / 376 (1.6%)
Wrong + uncertain	2
Correct + uncertain	4



Confidence Score: Correct vs Misclassified

374 correct predictions | 2 wrong predictions. Left: overlapping histogram shows wrong predictions cluster at lower confidence. Right: per-class boxplot isolates which orientation classes produce uncertain wrong predictions.



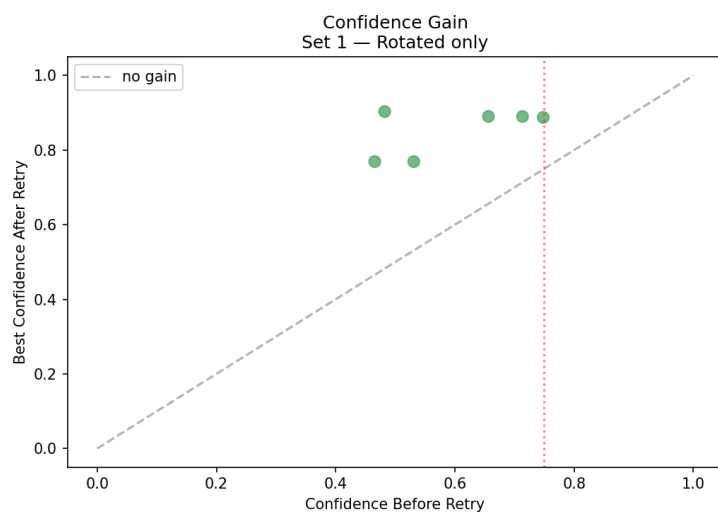
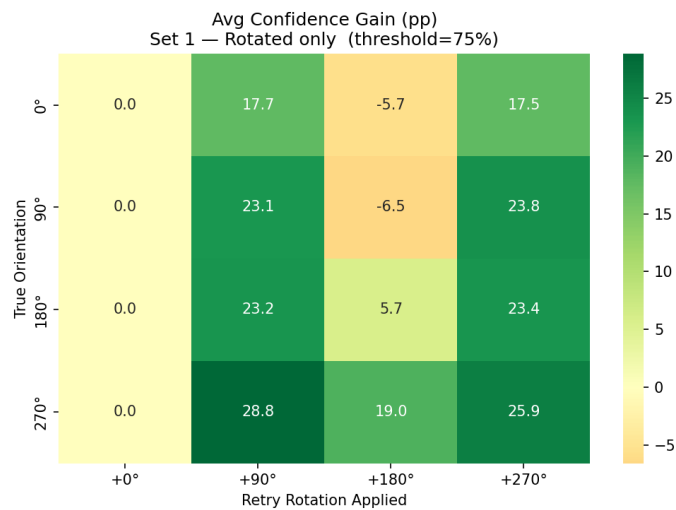
Retry Results:

Metric	Value
Uncertain images retried	6
Accuracy before retry	66.7%
Accuracy after retry	100.0%
Avg confidence gain	+25.2pp
Errors rescued	2
Correct flipped wrong	0

Avg confidence gain per retry rotation (pp):

True Deg	n	+0°	+90°	+180°	+270°
0°	1	+0.0pp (0%)	+17.7pp (100%)	-5.7pp (0%)	+17.5pp (100%)
90°	1	+0.0pp (0%)	+23.1pp (100%)	-6.5pp (0%)	+23.8pp (100%)
180°	1	+0.0pp (0%)	+23.2pp (100%)	+5.7pp (100%)	+23.4pp (100%)
270°	3	+0.0pp (0%)	+28.8pp (100%)	+19.0pp (100%)	+25.9pp (100%)

Green = strongest avg gain for that class.

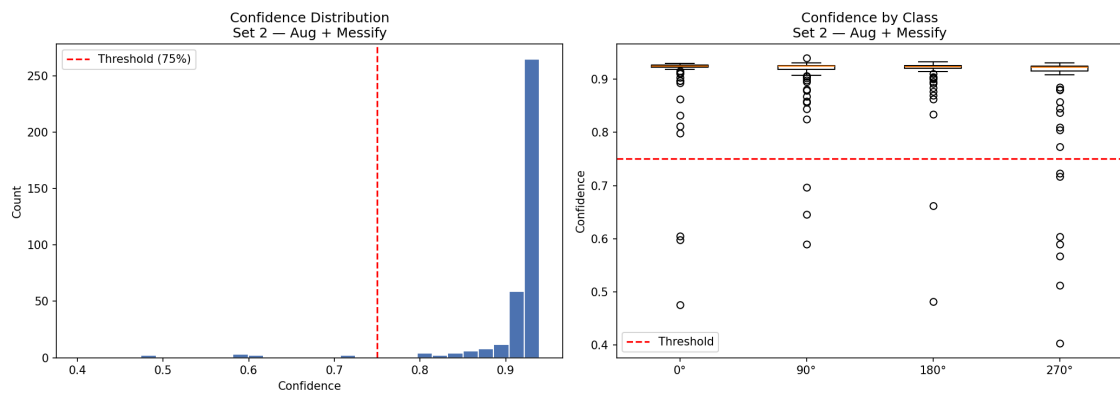


Rotate-to-0 Self-Consistency:

Metric	Value
Corrected → model says 0°	6 / 6 (100.0%)
When orig correct	100.0%
When orig wrong	100.0%

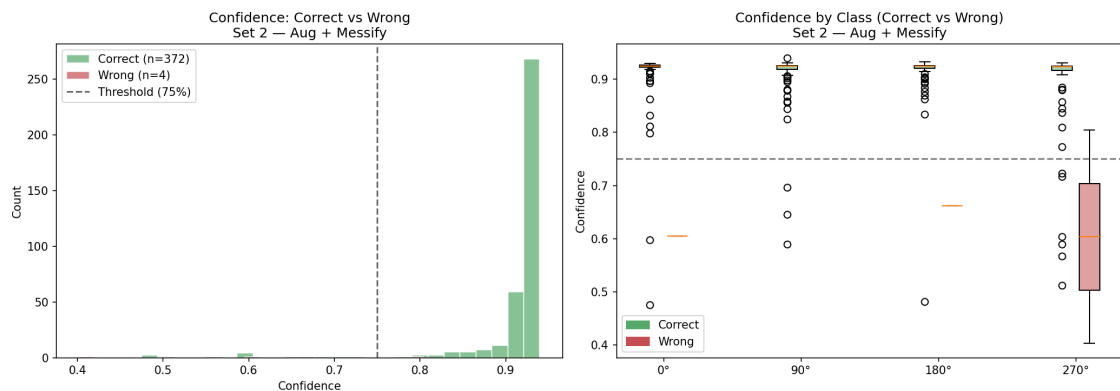
3. Set 2 — Aug + Messify

Metric	Value
Images	376
Overall accuracy	98.94%
Avg confidence	90.5%
Uncertain (<75%)	15 / 376 (4.0%)
Wrong + uncertain	3
Correct + uncertain	12
Clean accuracy	99.56%
Messy accuracy	98.00%



Confidence Score: Correct vs Misclassified

372 correct predictions | 4 wrong predictions. Left: overlapping histogram shows wrong predictions cluster at lower confidence. Right: per-class boxplot isolates which orientation classes produce uncertain wrong predictions.



Retry Results:

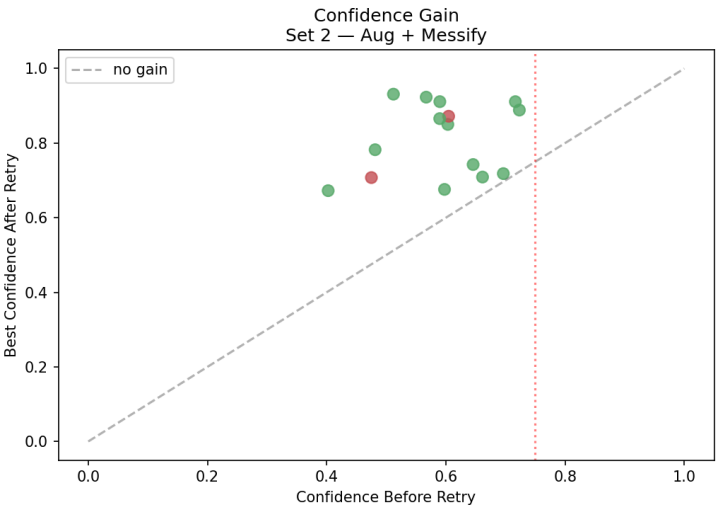
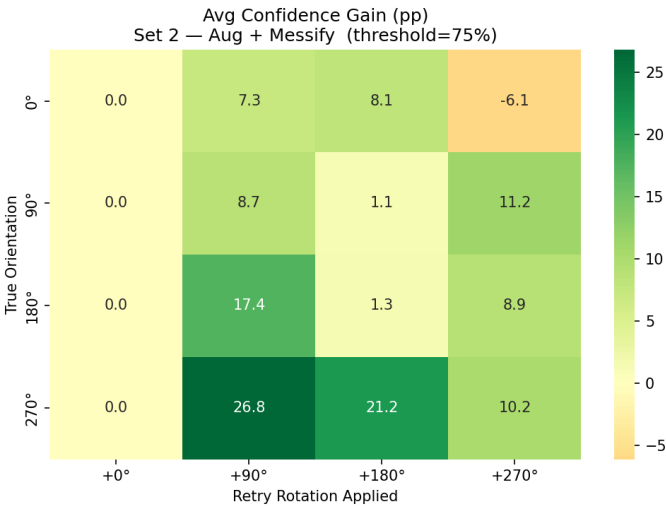
Metric	Value
Uncertain images retried	15
Accuracy before retry	80.0%
Accuracy after retry	86.7%

Avg confidence gain	+22.0pp
Errors rescued	2
Correct flipped wrong	1

Avg confidence gain per retry rotation (pp):

True Deg	n	+0°	+90°	+180°	+270°
0°	3	+0.0pp (0%)	+7.3pp (67%)	+8.1pp (67%)	-6.1pp (33%)
90°	3	+0.0pp (0%)	+8.7pp (67%)	+1.1pp (67%)	+11.2pp (67%)
180°	2	+0.0pp (0%)	+17.4pp (100%)	+1.3pp (50%)	+8.9pp (100%)
270°	7	+0.0pp (0%)	+26.8pp (100%)	+21.2pp (100%)	+10.2pp (71%)

Green = strongest avg gain for that class.



Rotate-to-0 Self-Consistency:

Metric	Value
Corrected → model says 0°	13 / 15 (86.7%)
When orig correct	100.0%
When orig wrong	33.3%

4. Set 1 vs Set 2 — Retry Comparison

Metric	Set 1 (Clean)	Set 2 (Aug + Messy)
Images	376	376
Overall accuracy	99.47%	98.94%
Avg confidence	91.5%	90.5%
Uncertain (<75%)	6 (1.6%)	15 (4.0%)
Retry acc (uncertain set)	100.0%	86.7%
Avg conf gain	+25.2pp	+22.0pp
Errors rescued	2	2

Model: PP-LCNet_x1_0_doc_ori | English-only, 7 PDFs | Threshold: 75% | Date: 2026-05-06